Electron states in mesa-etched one-dimensional quantum well wires

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Two-dimensional, self-consistent solutions of the Schrödinger and Poisson equations are used to find the electron states in GaAs/AlGaAs quantum well wires. Both deep and shallow mesa structures are simulated. Our results show that while these structures are capable of providing the single occupied subband and wide energy separations needed for a true quantum wire, the process tolerances allowed are very small, on the order of 200 Å of width variation. Cutoff widths calculated are 1000 Å for the shallow mesa and 2100 Å for the deep mesa. The agreement with experimental results is good for the shallow mesa, but poor for the deep mesa. This suggests additional process-induced sidewall depletion mechanisms contributing to the cutoff of the deep mesa structures.

I. INTRODUCTION

The reduced density of states to which electrons can scatter in "quantum structures" has led to a number of intriguing predictions of enhanced performance for electronic and optoelectronic devices. A wide variety of techniques have been employed and suggested to fabricate such structures.[1-5] One popular fabrication technique for quantum wires in the high-mobility III-V compound semiconductors is to use an epitaxial growth technique to form a two-dimensional electron gas (2DEG) at a single heterostructure or in a quantum well. This provides one dimension of electron confinement. The second dimension of confinement is provided by either etching through the 2DEG (deep mesa) or etching close to the 2DEG (shallow mesa). In both cases, depletion from the surface confines the electrons to a one-dimensional (1D) channel. The shallow mesa has been proposed as a means of eliminating surface recombination of carriers in the channel, and minimizing fabrication-related degradation at the sidewalls.[6] Another method of forming quantum wires is to use the depletion from biased Schottky gates to provide the second dimension of confinement.[5] In addition, these Schottky gates can be used to compensate for variations in structure size by adjusting the gate bias. Simulations of this quantum wire structure have been done by Laux, Frank, and Stern,[7] and are in a sense complementary to the self-consistent solutions of Schrödinger's and Poisson's equations that we describe here. We have carried out these calculations in two dimensions to more accurately model the energy levels and electron densities in the deep and shallow mesa quantum wires. The results show limits on the range of practical operation of electronic devices based on such structures, and indicate the fabrication tolerances required.

Various measurements made on 1D structures have verified the existence of quantized energy levels or subbands.[8] However, one theoretical promise of quantum wires, the higher electron mobility resulting from the reduced density of states to which the electrons can scatter,[9] has not been observed. This mobility enhancement depends on the strict limitation of scattering mechanisms, such as those induced by sidewall roughness or processing. An intrinsic limitation of mobility is set by interlevel scattering when the subband energy separation ΔE is on the order of kT. Our calculation highlights the low temperatures necessary to limit intersubband scattering. Finally, we show that the electron density, and hence the current carrying capability, is very low in the "quantum" regime. Therefore, it may be necessary to place arrays of quantum structures in parallel to achieve useful current densities, with the attendant problems of structure uniformity.

II. METHOD

The structures to be simulated are shown in Figs. 1(a) and 1(b). To find the conduction band, electron concentration, and electron states, it is necessary to solve both the Poisson and Schrödinger equations self-consistently. A numerical solution is needed, and the method of finite differences[10] is used in this work. The underlying equations are the two-dimensional Poisson equation:

$$\nabla\cdot\epsilon_s(x,y)\nabla\phi(x,y) = -q[N_D(x,y) - n(x,y)], \qquad (1)$$

where ϕ is the electrostatic potential, ϵ_s is the semiconductor dielectric constant, N_D is the ionized donor concentration, and n is the electron concentration; and the two-dimensional, time-independent Schrödinger equation:

$$-\frac{\hbar}{2}\nabla\cdot\frac{1}{m^*(x,y)}\nabla\psi(x,y) + V(x,y)\psi(x,y) = E\psi(x,y), \qquad (2)$$

where ψ is the wave function, E is the energy, V is the potential energy, $\hbar$ is Planck's constant divided by 2π, and m^* is the effective mass. In the structures presented here, all donors are ionized, and electron-electron interactions are neglected. The objective is to find a conduction-band distribution and an electron distribution which satisfy both of these equations. First, it is necessary to define how these physical quantities are related to the variables in Eqs. (1) and (2). The energy of the conduction-band edge is

$$E_c(x,y) = -q\phi(x,y) + \Delta E_c(x,y), \qquad (3)$$

where $\Delta E_c(x,y)$ is the pseudopotential due to the band offset between materials in the heterostructure, 0.23 eV for

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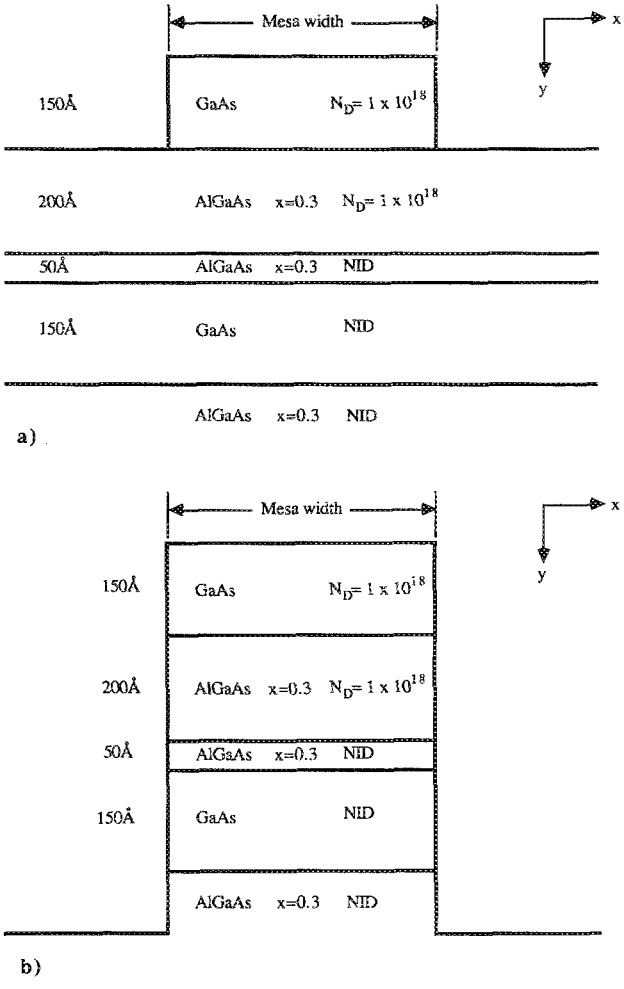

FIG. 1. (a) Schematic view of the shallow mesa structure. (b) Schematic view of the deep mesa structure.

$x = 0.3$ GaAs/AlGaAs. The potential term V in the Schrödinger equation is simply equal to the conduction-band energy. In these calculations thermal equilibrium is assumed, and the origin of all energies is chosen to be the Fermi level.

The energy eigenvalues of the Schrödinger equation give the quantized subband energies. These, in conjunction with the wave functions, are used to calculate the electron concentration in the structure by the following equation:

$$n(x,y) = \sum_{k=1}^{m} \psi_k^*(x,y)\psi_k(x,y) \times \int_{E_k}^{\infty} \frac{\sqrt{2m^*}}{\pi\hbar\sqrt{E-E_k}} \frac{dE}{[1+\exp(E-E_f/kT)]}, \quad (4)$$

where the summation runs over all the subbands, and the integral includes the 1D density of states and the occupation probability.

The numerical solutions of Eqs. (1) and (2) are boundary value problems. Poisson's equation is solved over the entire structure, where the free-surface boundary conditions are a surface barrier height set by Fermi-level pinning. The values used are $E_c = 0.6$ eV at the surface of GaAs and 0.7 eV at the surface of AlGaAs. These values were chosen based on experiments with 2DEG structures. Schrödinger's equation is solved in the limited region of the structure which contains electrons. The wave function ψ must be zero at the boundaries of this region. The finite difference formulation of Schrödinger's equation gives a matrix which is sparse, real, and symmetric. The eigenvalues and eigenvectors of this matrix are the solution of the Schrödinger equation. Matrix symmetry, which holds for uniform mesh spacings, greatly simplifies the solution. Accordingly, uniform mesh sizes are used. Typical mesh sizes are $\Delta x = 20$ Å and $\Delta y = 10$ Å. The EISPACK[11] routines are used to find the eigensolutions. Equation (4) uses the solution of Schrödinger's equation to calculate an electron concentration, which is then used in finding a solution to Poisson's equation, related to the conduction band by Eq. (3). The finite difference form of Poisson's equation produces a system of linear equations and unknowns. This system, which for the problems we are considering can be as large as 10 000 equations, is solved using a matrix overrelaxation technique. Self-consistent convergence of the solutions of both equations is imposed using the Newton–Raphson method.[12] Our simulations were run on a Sun-4 workstation, with typical run times of 4–6 h.

III. RESULTS

Figures 2 (a) and 2(b) show the conduction band and electron distribution resulting from our calculations, along a slice taken through the middle of a shallow mesa structure, from surface to substrate. The width of the mesa is 1500 Å, and the simulation temperature is 4 K. The peak of the electron distribution is located 50 Å from the edge of the well, confined by the AlGaAs barrier layers. Figure 3 (a) shows the conduction band for three shallow mesa widths, along a slice in the x direction, taken at $y = 410$ Å, just inside the quantum well. These conduction bands show some interesting characteristics. In the shallow mesa structure, more than 200 meV of confinement is provided by surface depletion in the etched regions. Also, the conduction band flattens out approximately 500 Å inside the edge of the widest mesa. If the mesa is about 1000 Å wide, the effects of the side depletion meet in the middle, and the bottom of the band will be parabolic in shape. Below 1000 Å, the bottom of the well will be pulled away from the Fermi level. Eventually the electron states will be so far from the Fermi level that they will have little probability of being occupied, and the wire conductivity is "cutoff." The same is true for the deep mesa, but the cutoff occurs at wider widths, as shown in Fig. 3(b).

As the width of the mesa is decreased, the energies of the bound states increase, and the energy difference between these subbands also increases. Figure 4 shows the occupation probability of the lowest state of the first and second subbands as a function of mesa width at 4 K. Figure 5 shows the energy separation between these subbands at 4 K, in meV and referenced to kT, as a function of mesa width. To limit interlevel scattering, an ideal quantum wire structure will have subband separations of several kT, with

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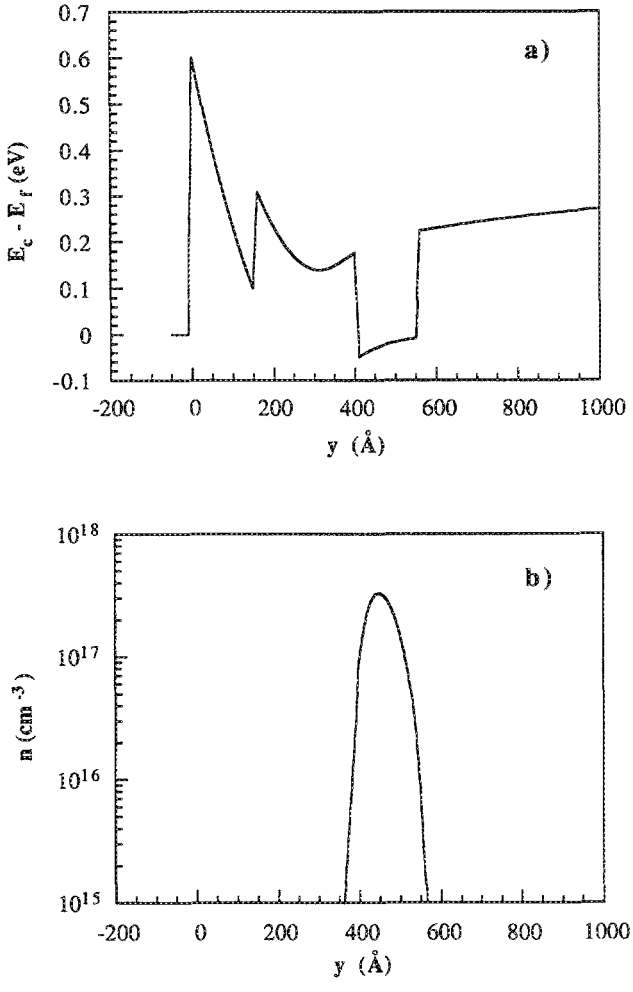

FIG. 2. (a) Energy-band diagram taken at the middle of the mesa, from surface to substrate, for the shallow etched structure of Fig. 1(a). (b) The corresponding electron concentration.

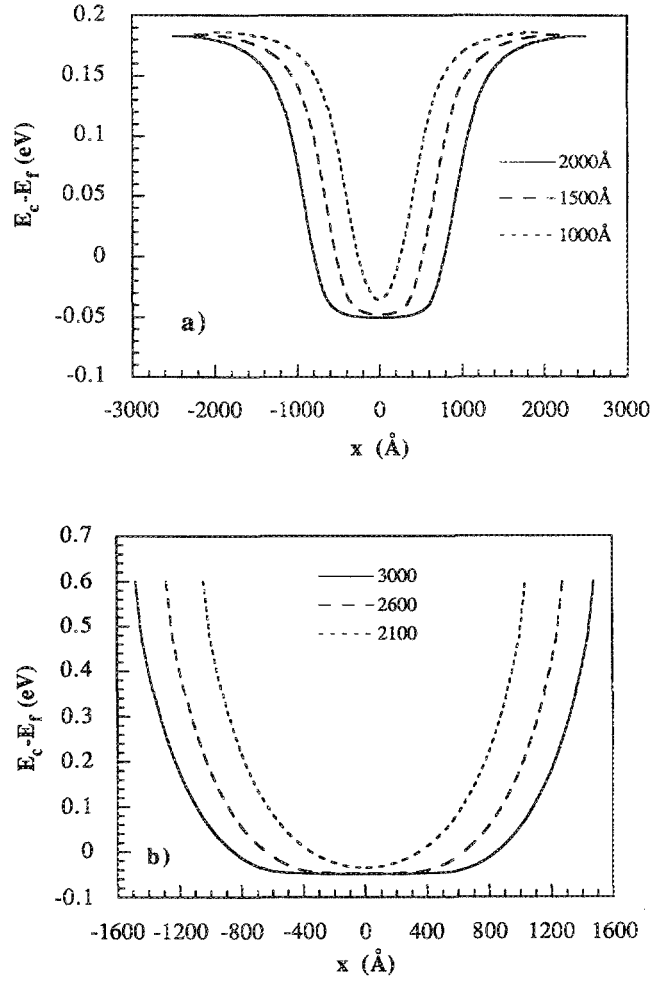

FIG. 3. (a) Band diagram for the shallow mesa in the x direction, inside the quantum well. Three mesa widths are shown: 1000, 1500, and 2000 Å. (b) Band diagram for the deep mesa with widths of 2100, 2600, and 3000 Å.

only the first subband occupied. This limits the interlevel scattering, and the structure will behave like a "true quantum wire." These goals are difficult to achieve in mesa-etched structures since the wire is very close to cutoff when the state separation is sufficiently large. At 4 K, we see from Fig. 4 that these conditions can be met with a shallow mesa width between 1000 and 1200 Å, allowing little room for process variation. Lowering the temperature further does not increase the process latitude, since the first subband energy is below the Fermi level for widths greater than 1000 Å, and the second subband energy is below the Fermi level for widths greater than 1200 Å, even at low temperatures. The deep mesa structure also has a 200-Å window, from 2200 to 2400 Å, where the requirements of a quantum wire are satisfied. The subband spacings shown in Fig. 5 are about 10 meV near cutoff, which is larger than has been observed experimentally.[8,13] One possible explanation for the discrepancy is the presence of parallel conduction channels in the barrier layers of the experimental structures. Depending on the material structure, the subband energies associated with these channels can be closely spaced, and can also be close in energy to the intentional

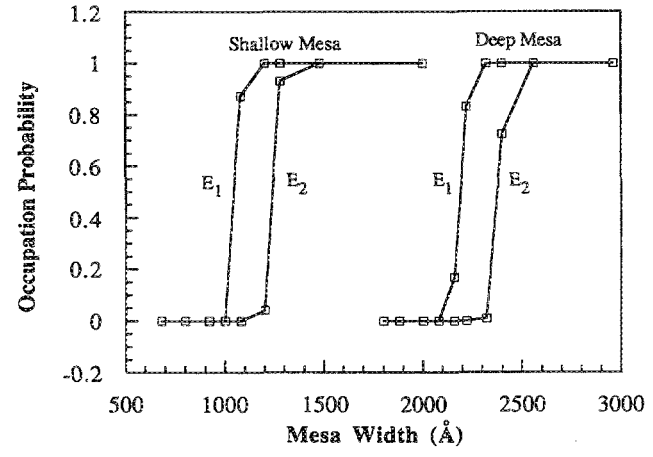

FIG. 4. Occupation probability of the lowest state in the two lowest subbands E_1 and E_2, plotted as a function of mesa width, for both the deep and shallow mesas at 4 K.

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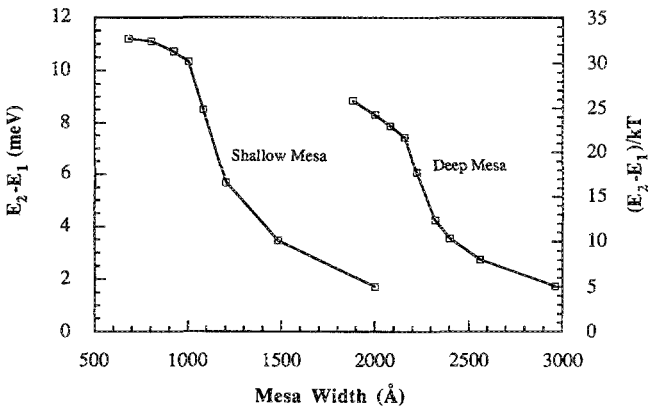

FIG. 5. Separation of the first two subbands, E_2-E_1, for both structures at 4 K. The right axis is normalized to kT.

subbands at the heterojunction. Therefore, to achieve large subband spacing it is vitally important to eliminate parallel conduction channels.

Below 1000 Å for the shallow mesa and 2200 Å for the deep mesa, the conduction band in the x direction is parabolic at the bottom of the wire, and the lower-energy levels become evenly spaced, as in a harmonic oscillator. In this harmonic oscillator region, the energy-level spacings are not a strong function of mesa width, and narrowing the mesa further causes only a very small increase in energy separation. So it is evident that the limiting factor in these structures is the nonabrupt confinement provided by the surface depletion.

As the width of the mesa is reduced, the number of electrons in the wire per unit length, n_l, will also decrease. Figure 6 shows the linear electron concentration as a function of width for two temperatures, 77 and 4 K. In a simple mobility model (nonballistic), the conductance of a wire will differ from the data in Fig. 6 by a multiplicative factor. This means that at 4 K, wires formed by a shallow mesa etch should exhibit a cutoff in conductivity at about 1000 Å, and 2100 Å for the deep mesa. Measurements made by other investigators indicate a cutoff of about 900 Å for the shallow mesa[14] and approximately 4000Å for the deep

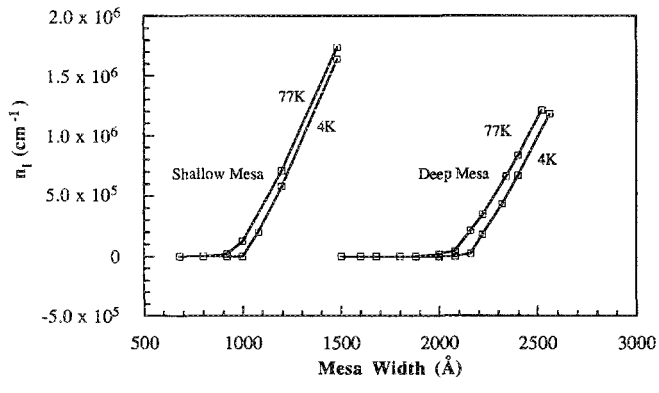

FIG. 6. Linear electron concentration n_l as a function of mesa width for both structures. Data for both 4 and 77 K are shown.

mesa structure.[8,15] The discrepancy for the deep mesa suggests that some mechanism in addition to surface depletion plays a role in wire cutoff. Some possibilities are enhanced surface depletion due to damage caused by dry etching, or donor passivation.[16–18]

A subtle feature of the data shown in Fig. 6, most evident for the shallow mesa, is the behavior of the linear electron concentration as it approaches zero. For widths greater than 1000 Å for the shallow and 2200 Å for the deep mesa, any decrease in mesa width merely removes part of the flat portion of the conduction band shown in Fig. 3, and the decrease of the electron concentration is linear. Below these widths, the conduction band in the x direction approximates a parabola. As seen before, the state separations in this regime are not strongly dependent on the mesa width. At 77 K, several subbands are occupied, while at 4 K only one or two subbands are occupied. Therefore, at 77 K, in this parabolic regime, there is a decrease in the slope of n_l versus width, since the upper subbands are not being depopulated as rapidly as at greater widths. This effect also occurs at 4 K, but over a smaller width range. The result is more electrons in the wire than a straight-line extrapolation of the linear region of n_l would predict. Since one manifestation of enhanced 1D mobility would be greater than expected conductance near cutoff, care should be taken when making conductance measurements not to mistake this decrease in slope for enhanced mobility. However, the effect occurs over a very small width range and near cutoff making it very hard to detect.

IV. CONCLUSION

Self-consistent solutions to the Schrödinger and Poisson equations for both shallow and deep etched mesa structures have been calculated. These simulations predict a wire cutoff width of 1000 Å for the shallow mesa and 2100 Å for the deep mesa. Widths below these values only result in a loss of conduction as the electron states move away from the Fermi level. Measurements performed by other investigators indicate good agreement in cutoff width for the shallow mesa. Data taken for a deep mesa do not agree well with the results of these simulations, which implies that more than simple surface depletion is contributing to wire cutoff. Mechanisms suggested include dry etch damage and donor passivation. Future work in these simulations should include models for these process-induced effects. The simulations we have performed, in conjunction with measurements of fabricated structures, should provide a powerful tool for studying the effects of dry etching.

Our simulations show that both the shallow and deep mesa structures are capable of meeting the goals of a quantum wire: one occupied subband with large subband separation. However, our simulations suggest that it may be difficult to observe quantum wire behavior in these structures. It is possible to achieve the goals, but the allowable width variation is only 200 Å. The process control necessary to achieve this variation over long wires is prohibitive. If the requirement of one occupied state is relaxed, the allowable width variation increases at the cost of increased interlevel scattering. A subtle result of quantum confine-

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ment is the decrease in the slope of n_l near cutoff at higher temperatures. This effect occurs over a narrow range of widths near cutoff, and again fabrication tolerances may make this difficult to observe.

These self-consistent solutions of the Schrödinger and Poisson equations provide an accurate model of the energy levels and electron densities in deep and shallow mesa quantum wires. Our results show that these structures place very stringent requirements on fabrication to provide the small variations in width needed for quantum wire behavior. Additional work is needed to more accurately model the free surfaces of the semiconductor, in particular the etched sidewalls of the deep mesa.

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